MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)
THIRD SEMESTER EXAMINATION (2024 ADMISSION ONWARDS)

MG3DSCSTA201 – Analytical Tools for Multivariate Analysis

Duration: 1Hour Maximum: 30 Marks

PRACTICAL EXAMINATION

Answer any five questions. Each question carries 6 marks.

Solve the questions using R software in computer lab. The data in the question along with their answer must be written in the answer paper.

1. Examine linear independence of the following vectors [1,2, -1], [2,1,4], [1,7,17] [A][1]
2. Check whether the given matrix is orthogonal or not $\begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$ [A][1]
3. Compute the determinant of the matrix $\begin{bmatrix} 67 & 19 & 21 \\ 39 & 13 & 14 \\ 81 & 24 & 26 \end{bmatrix}$ [A][2]
4. Find the inverse of the matrix $\begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 3 & 1 & 1 \end{bmatrix}$ [A][2]
5. Find rank of the matrix $\begin{bmatrix} 2 & 3 & 4 \\ 3 & 1 & 2 \\ -1 & 2 & 2 \end{bmatrix}$ [A][2]
6. Solve the system of equations: x+3y-2z=0
   2x-y+4z=0
   x-11y+14z=0 [A][3]

7. Obtain g inverse of the matrix $\begin{bmatrix} 1 & -1 & 2 \\ 2 & 6 & 3 \\ 3 & 13 & 4 \end{bmatrix}$ [A][4]
8. Determine characteristic roots and corresponding vectors of the matrix $\begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$ [A][4]